

MARITIME INTELLIGENCE

SEADRAGON

Maritime Data Platform

We turn live ship signals into clear operational insight — so leaders can see what is happening at sea, without needing to speak database.

One shared data backbone.
Many applications on top.
Always monitored. Always backed up.

In this briefing

The shared AIS backbone

How ship signals become trusted working data.

Applications that reuse it

JKPTG, TSS Reporting, and maritime intelligence apps.

What the system can spot

Going dark, close ship pairs, sensitive anchoring — in plain language.

Operations you can trust

Grafana monitoring and AWS S3 retention.

Why This Platform Exists

Shipping generates continuous AIS messages — identity, position, speed and status. Turning that flood into **decisions** is hard: the data never stops, volumes are large, and meaningful events are rare.

What managers usually feel

Too many hand-offs

Collection, storage and reporting often live in separate tools.

Insight arrives late

Or not in a form the operations room can use.

Stacks are costly to keep

Open-source big-data platforms need constant care.

Apps rebuild plumbing

Every new product risks reinventing AIS collection.

Our answer in one line

Make live maritime analytics as straightforward as **collect** → **understand** → **act**.



Collect

Receive AIS continuously from the JLM source.



Understand

Python pipeline decodes and prepares every message.



Act

Apps and APIs deliver maps, reports and alerts.

Seadragon standardises the hard middle — so product teams focus on business outcomes, not reinventing data plumbing.

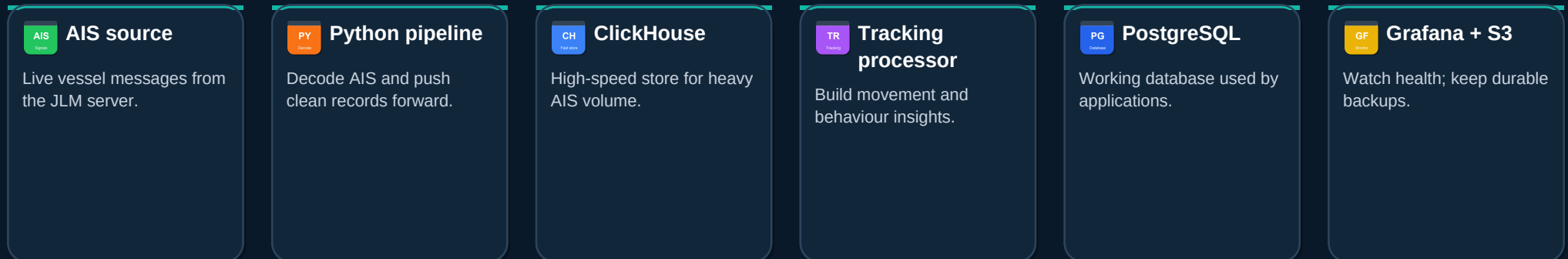
The Shared Data Backbone

Every major maritime app in this estate starts from the same reliable path. Think of it as the harbour road that all terminals share.

Shared Maritime Data Backbone



Monitored with Grafana · Backed up & retained on AWS S3



This core structure is what JKPTG and TSS share — ingest, decode, fast store, tracking, then PostgreSQL for day-to-day use.

From Tracking to Intelligence

Beyond “where is the ship?”, the same data can answer “is something unusual happening?”. The PySTS / MANTIS layer turns movement history into practical alerts.

Intelligence on the same pipeline (MANTIS / PySTS)



**Going dark**

A ship slows, then AIS goes quiet before a normal stop is confirmed. Useful for security attention — labelled as suspected, not proven intent.

**Ship-to-ship closeness**

Cargo and tankers that remain unusually close are scored by duration and whether the mix looks like a transfer pattern.

**Anchoring watch**

Large commercial ships stopped inside restricted or watch waters — busy port traffic filtered so the signal stays clear.

Delivered through secure MANTIS APIs and map views — ready for briefings, not just engineers.

Always On: Monitor & Protect

A data platform is only as good as its day-to-day reliability.



Grafana monitoring

Live dashboards watch pipeline health — so delays, failures or overload are seen early, not after a report goes missing.



AWS S3 backup & retention

Python jobs copy and retain critical data in AWS S3 — durable storage for recovery, audit and long-term history.

What this means for management

Continuity

Applications stay fed even when one consumer changes.

Accountability

Monitoring shows whether the pipeline is healthy today.

Durability

S3 retention protects against loss and supports investigations.

Reuse

New apps plug into PostgreSQL / APIs instead of re-ingesting AIS.

In short: the estate is designed to be **operated**, not only built once.

Proven Under Load

The platform approach has already been exercised at serious maritime volumes.

9M+

AIS messages processed in a single stress-test day

20M+

Derived insights generated for deeper analysis

Live

Pipeline monitored with Grafana

Shared

One backbone serving multiple applications

Example outcome: the same testing stream supported real-time GHG-style reporting aligned with IMO guidance — showing the backbone can feed both **operations** and **compliance** narratives.

PG
PostgreSQL

Working data

PostgreSQL holds tracking and app-ready tables.

API
REST

Programmatic access

REST APIs for dashboards and partner systems.

STS
Ships

Behaviour insight

Proximity, slow-down and dark-candidate logic.

AN
Anchorage

Zone awareness

Restricted and anchorage watch overlays.

Who Benefits

Operations & coast watch

Faster situational picture — where ships are, and which clusters need attention.

Compliance & reporting teams

Consistent source data for GHG, traffic and programme reporting.

Product & system teams

Reuse the backbone; spend effort on user experience and rules, not AIS plumbing.

Leadership

One architecture story: collect once, serve many, monitor and protect always.

What You Get

Capability package in plain terms

Near real-time ingestion

AIS from the operational JLM source.

Decoded, query-ready stores

ClickHouse for volume, PostgreSQL for apps.

Tracking processors

Maintain vessel movement context continuously.

Application pipelines

JKPTG, TSS Reporting and intelligence APIs.

Dashboards & maps

Human briefings without raw database work.

Monitoring & backup

Grafana visibility, AWS S3 retention.

Flexible delivery

API-first so partners and internal apps can connect.

Roadmap Direction

Priorities stay practical: strengthen the backbone, widen insight, keep operations calm.

Now

Stable AIS backbone, tracking into PostgreSQL, Grafana + S3, live apps (JKPTG / TSS / MANTIS APIs).

Next

Clearer dark-vessel confidence, coverage-aware filters, richer briefing dashboards, tighter API packaging.

Later

Deeper analytics for programme owners, more automated alerts, broader partner access with strong tenancy controls.

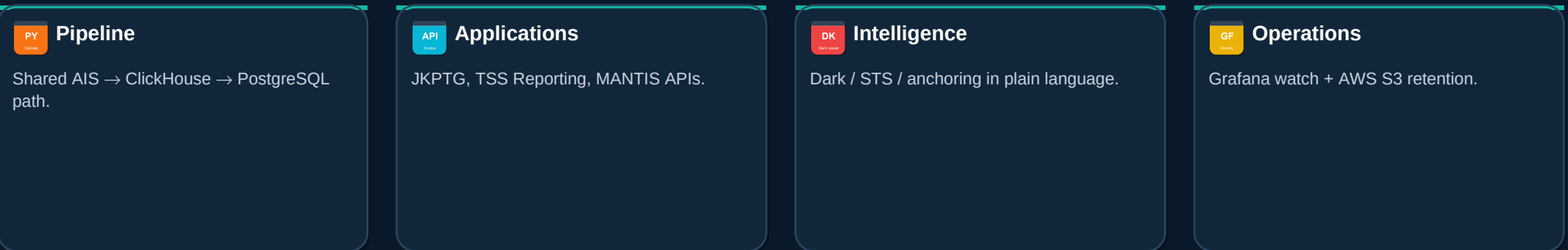
How to talk about it externally

One-sentence story

"We run a shared maritime data platform: AIS is collected and decoded once, stored for speed and for applications, enriched by tracking processors, monitored in Grafana, retained on AWS S3, and consumed by multiple programmes through databases and APIs — including intelligence for dark activity, close-quarters pairs and sensitive anchoring."

Ready for the next conversation?

Show the backbone once. Then show how JKPTG, TSS Reporting and MANTIS each ride it. Finish with monitoring and backup — so the story feels complete to management.



Request a live walkthrough
Pipeline · Applications · Intelligence · Operations

Seadragon Maritime Data Platform · Support local, serve the region

Intelligence outputs are operational aids. Combine with local rules, permits and human review.